

1 Grammar Rules

Types and Contexts

$$\begin{array}{ll}
\sigma, \tau & ::= \mathbf{unit} \mid \mathbf{num} \mid \sigma \otimes \sigma \mid \alpha \quad (\text{Types}) \\
\alpha & ::= m(\sigma) \quad (\text{Discrete Types}) \\
\Gamma & ::= \emptyset \mid \Gamma, x :_r \sigma \quad (\text{Linear Typing Contexts}) \\
\Phi & ::= \emptyset \mid \Phi, z : \alpha \quad (\text{Discrete Typing Contexts})
\end{array}$$

Expressions

$$\begin{array}{lcl}
e, f & ::= & x \mid z \mid () \mid !e \mid (e, f) \mid \mathbf{inl} e \mid \mathbf{inr} e \\
& | & \mathbf{let} x = e \mathbf{in} f \mid \mathbf{let} (x, y) = e \mathbf{in} f \mid \mathbf{dlet} z = e \mathbf{in} f \mid \mathbf{dlet} (x, y) = e \mathbf{in} f \\
& | & \mathbf{case} e' \mathbf{of} (\mathbf{inl} x.e \mid \mathbf{inr} y.f) \mid \mathbf{add} e f \mid \mathbf{sub} e f \mid \mathbf{mul} e f \mid \mathbf{dmul} e f \mid \mathbf{div} e f
\end{array}$$

2 Typing Rules

Context Variables

$$\frac{}{\Phi \mid \Gamma, x :_r \sigma \vdash x : \sigma} (\text{Var}) \qquad \frac{}{\Phi, z : \alpha \mid \Gamma \vdash z : \alpha} (\text{DVar})$$

Tensor Destructors

$$\frac{\Phi \mid \Gamma \vdash e : \tau_1 \otimes \tau_2 \quad \Phi \mid \Delta, x :_r \tau_1, y :_r \tau_2 \vdash f : \sigma}{\Phi \mid r + \Gamma, \Delta \vdash \mathbf{let} (x, y) = e \mathbf{in} f : \sigma} (\otimes E_\sigma)$$

$$\frac{\Phi \mid \Gamma \vdash e : \alpha_1 \otimes \alpha_2 \quad \Phi, z_1 : \alpha_1, z_2 : \alpha_2 \mid \Delta \vdash f : \sigma}{\Phi \mid \Gamma, \Delta \vdash \mathbf{dlet} (z_1, z_2) = e \mathbf{in} f : \sigma} (\otimes E_\alpha)$$

Let Bindings

$$\frac{\Phi \mid \Gamma \vdash e : \tau \quad \Phi \mid \Delta, x :_r \tau \vdash f : \sigma}{\Phi \mid r + \Gamma, \Delta \vdash \mathbf{let} x = e \mathbf{in} f : \sigma} (\text{Let}) \qquad \frac{\Phi \mid \Gamma \vdash e : \alpha \quad \Phi, z : \alpha \mid \Delta \vdash f : \sigma}{\Phi \mid \Gamma, \Delta \vdash \mathbf{dlet} z = e \mathbf{in} f : \sigma} (\text{DLet})$$

Bang Destructor

$$\frac{\Phi \mid \Gamma \vdash e : \sigma}{\Phi \mid \Gamma \vdash !e : m(\sigma)} (\text{Disc})$$

Operations

$$\frac{}{\Phi \mid \Gamma, x :_{\varepsilon+r_1} \mathbf{num}, y :_{\varepsilon+r_2} \mathbf{num} \vdash \{\mathbf{add}, \mathbf{sub}\} x y : \mathbf{num}} (\text{Add, Sub})$$

$$\frac{}{\Phi \mid \Gamma, x :_{\varepsilon/2+r_1} \mathbf{num}, y :_{\varepsilon/2+r_2} \mathbf{num} \vdash \mathbf{mul} x y : \mathbf{num}} (\text{Mul})$$

$$\frac{}{\Phi \mid \Gamma, x :_{\varepsilon/2+r_1} \mathbf{num}, y :_{\varepsilon/2+r_2} \mathbf{num} \vdash \mathbf{div} x y : \mathbf{num} + \mathbf{unit}} (\text{Div})$$

$$\frac{}{\Phi, z : m(\mathbf{num}) \mid \Gamma, x :_{\varepsilon+r} \mathbf{num} \vdash \mathbf{dmul} z x : \mathbf{num}} (\text{DMul})$$

References

- [1] Ariel E. Kellison, Laura Zielinski, David Bindel, and Justin Hsu. Bean: A language for backward error analysis. *Proceedings of the ACM on Programming Languages*, 9(PLDI):1838–1862, 2025.